

Wikipedia Article Generation - Island

Group 3

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Outline

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 - Evaluation Methodology
 - Error Analysis Methodolgy
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The Problem with Existing Approaches

STORM (Shao et al., 2024)

Iterates web queries to build outlines — but lacks modeling of **dependencies between sections**.

WikiGenBench (Zhang et al., 2024)

Plan-Retrieve-Read (PRR) pipeline causes "Information Silos" — **redundancy and incoherence across sections**. ¹

Static Data Reliance ²

Existing works rely on fixed datasets, missing real-time updates and recent events.



Our Proposed Solution

Main Goal: Break the Silos

Inter-section aware RAG—model dependencies across sections to produce a unified, coherent narrative.

Secondary Goal: Eliminate Static Data

Tavily Search API—Replace fixed datasets with a **fully automated, real-time retrieval pipeline**.

Current status

The island remains uninhabited, with empty high-rise buildings, seawalls and other mine-era structures. Contemporary travel writing and tourism media commonly present Hashima as an abandoned destination associated with heritage tourism, photography and urban exploration. Its ruins have inspired academic and creative work, including the AHRC-linked project 'The Future of Ruins: Reclaiming Hashima Island' and the related 2013 performance 'Return to Battleship Island'.

Hashima is widely circulated through popular culture and digital media, including fictionalized survival-horror games. A survey of immersive and virtual-reality projects found that such projects often ignore the former community life of the island or the visual spectacle of decay, while tending to avoid controversy. Educational and heritage-oriented digital resources have used Hashima to examine contested heritage memory politics and commercial tourism.

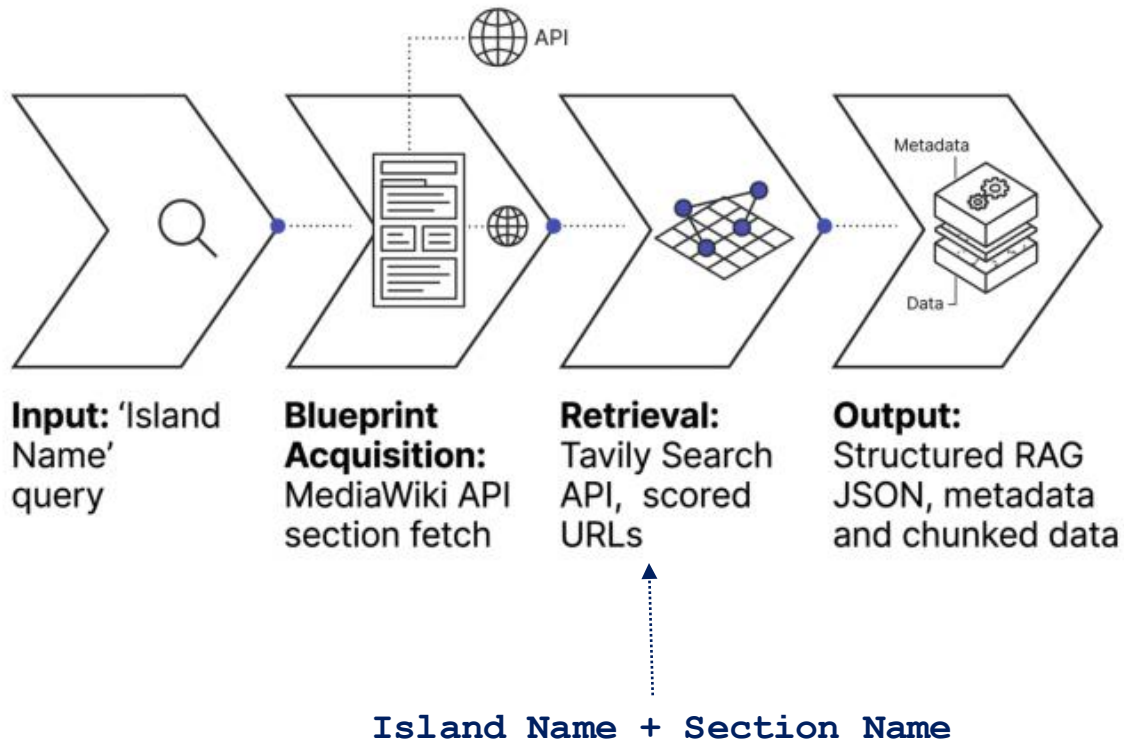
The island's interpretation continues to be the subject of diplomatic disagreement. At the 4th World Heritage Committee in Paris, South Korea sought renewed examination of Japan's follow-up to the 1993 agreement. Japan argued that it had implemented earlier resolutions. The committee backed the position that the issue should be addressed through bilateral dialogue rather than UNESCO procedures. Japan's revised proposal was rejected by ballot, with seven votes in favor, three against and several abstentions among the 21 committee members. The outcome, further formal discussion of the issue during the session, or in future committee meeting.

References

- [1] https://www.uwosh.edu/faculty_staff/earns/hashima.html
- [2] <https://reachtv.com/news/asias-forgotten-territories/>
- [3] <https://www.historicalgames.net/island-that-will-not-load/>
- [4] <https://gtr.ukri.org/projects?ref=AH%2FK005308%2F1>
- [5] <https://japan-forward.com/unesco-backs-japans-position-on-hashima-coal-mine-dispute/>



Retrieval Data Pipeline





Dynamic Retrieval & Chunking Rules

Hard Filter

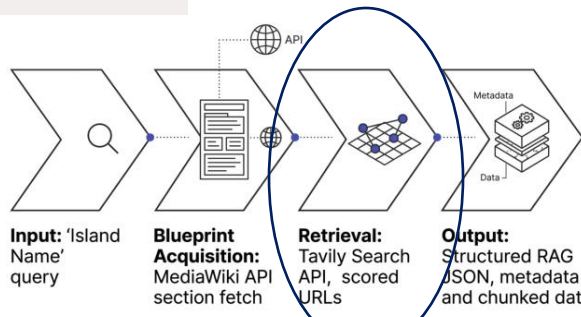
- **10** retrieved web contents
- Exclude **Wikipedia Link**
- Exclude Retrieval web data **fewer than 100 characters**
- Exclude **excessive external links**

Dynamic Retention

- **Score threshold:** Retain chunks with Tavily score > 0.65
- **Minimum guarantee:** Always keep Top 5 scoring chunks

Chunk Size

- **Chunk size:** 256 words (*Borgeaud et al., 2022*)
- **Overlap:** 60-words
- **Residual discard:** Fragments under 50 characters are dropped





Retrieval

Retrieval Data Output

3* Metadata Layer

Field	Description
original_input	Raw query (e.g., "Hunga Tonga")
resolved_entity_name	Disambiguated entity name
total_sections	Count of valid sections found

10* Islands



Surtsey

Volcanic origin



Hunga Tonga

2022 eruption



Hashima

Abandoned industry



Tromelin

Remote territory

6* Blueprint Data Layer

Field	Description
island_name	Target geographic entity
sections_data	Nested dict by section heading
chunk_id	Unique fragment identifier
text	256-word retrieved content
<u>source_url</u>	URL
<u>similarity_score</u>	<u>Tavily</u> Score



Four Article Generation Methods

- Same input: pre-built JSON with section headings and retrieved chunks
- Incremental design: each method adds one structural element over the previous

Baseline

Method 0: Pure Generation

- Generate full article at once
- No chunks used

WikiGEN

Method 1: Naive RAG

- WikiGenBench's Retrieve-then-Read (RR)
- Generate full article at once

WikiGEN

Method 2: Hierarchical RAG

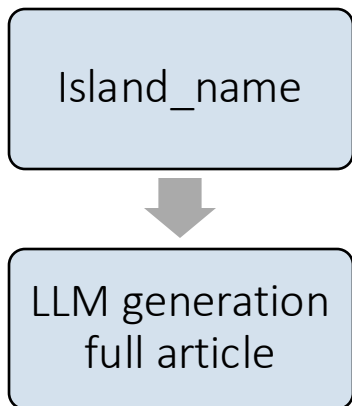
- WikiGenBench's Plan-Retrieve-Read (PRR)
- Generate section-by-section

Method 3: Inter-Section Aware RAG

- **Our proposed method**
- Extends Method 2 with an explicit DAG-based dependency mechanism, propagating generated-section summaries as context for downstream sections.



Method 0: Pure Generation -Baseline



```
SYSTEM_PROMPT = (
```

```
"You are an expert encyclopedia writer. "
```

```
"Write accurate, well-structured Wikipedia-style articles."
```

```
)
```

```
ARTICLE_PROMPT = """\
```

```
Write a full Wikipedia article about the following island: {island_name}
```

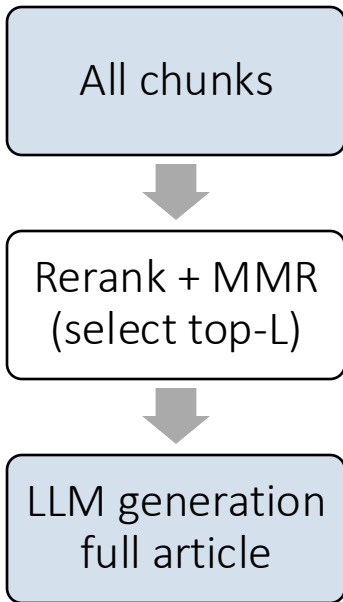
Requirements:

- Start directly with the first section, do not write a lead paragraph
 - Organize the article into sections
 - Start each section with ==SECTION NAME==
 - Do NOT use sub-sections (===Sub-section===) — use only top-level sections
 - Do NOT include a "References", "See also", or "External links" section at the end
 - Write in a neutral, encyclopedic tone
 - Cover geography, geology, ecology, climate, and history where relevant
- ```
"""
```



# Method 1: Naive RAG

(WikiGenBench: Retrieve-then-Read)



```
SYSTEM_PROMPT = (
 "You are an expert encyclopedia writer. "
 "Write accurate, well-structured Wikipedia-style articles with citations."
)
```

```
ARTICLE_PROMPT = """\n
I have a topic "{island_name}" that contains the following documents:
```

```
{documents}
```

Based on the above information, write a Wikipedia article about this island.

Start directly with the first section, do not write a lead paragraph.

Organize the content by sections. Before writing each section, always start with  
"==SECTION NAME==".

You MUST cite the most relevant document for every sentence you write.

Use ONLY this exact citation format: "This is an example sentence.[1]" or "Another  
sentence.[2][3]"

where the number inside the brackets refers to the Document number above.

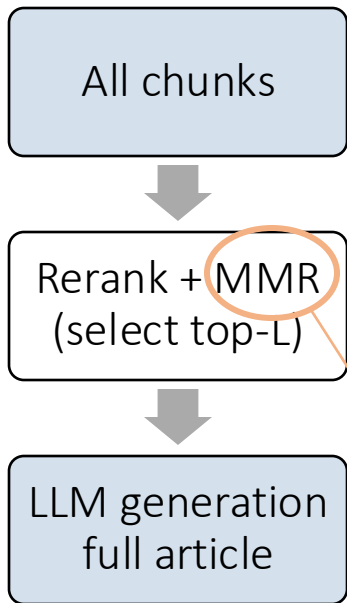
Do NOT write a "References", "See also", or "External links" section at the end —  
only inline citations like [1], [2], [3].

```
"""
```



# Method 1: Naive RAG

(WikiGenBench: Retrieve-then-Read)



Selects the most relevant chunks while avoiding repetition.

```
SYSTEM_PROMPT = (
 "You are an expert encyclopedia writer. "
 "Write accurate, well-structured Wikipedia-style articles with citations."
)

ARTICLE_PROMPT = """\n
I have a topic "{island_name}" that contains the following documents:

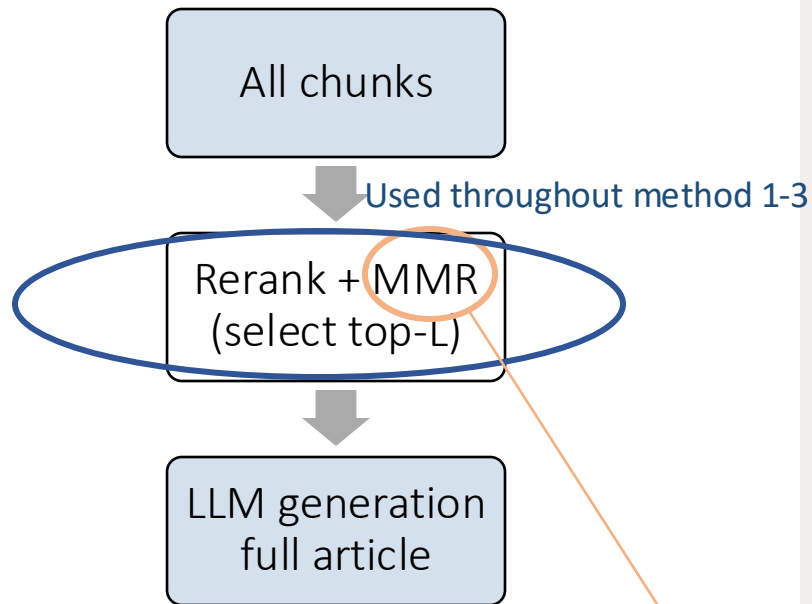
{documents}

Based on the above information, write a Wikipedia article about this island.
Start directly with the first section, do not write a lead paragraph.
Organize the content by sections. Before writing each section, always start with
"==SECTION NAME==".
You MUST cite the most relevant document for every sentence you write.
Use ONLY this exact citation format: "This is an example sentence.[1]" or "Another
sentence.[2][3]"
where the number inside the brackets refers to the Document number above.
Do NOT write a "References", "See also", or "External links" section at the end —
only inline citations like [1], [2], [3].
"""
```



# Method 1: Naive RAG

(WikiGenBench: Retrieve-then-Read)



```
SYSTEM_PROMPT = (
 "You are an expert encyclopedia writer. "
 "Write accurate, well-structured Wikipedia-style articles with citations."
)
```

```
ARTICLE_PROMPT = """\n
I have a topic "{island_name}" that contains the following documents:
```

```
{documents}
```

```
Based on the above information, write a Wikipedia article about this island.
```

```
Start directly with the first section, do not write a lead paragraph.
```

```
Organize the content by sections. Before writing each section, always start with
"==SECTION NAME==".
```

```
You MUST cite the most relevant document for every sentence you write.
```

```
Use ONLY this exact citation format: "This is an example sentence.[1]" or "Another
sentence.[2][3]"
```

```
where the number inside the brackets refers to the Document number above.
```

```
Do NOT write a "References", "See also", or "External links" section at the end —
only inline citations like [1], [2], [3].
```

```
"""
```

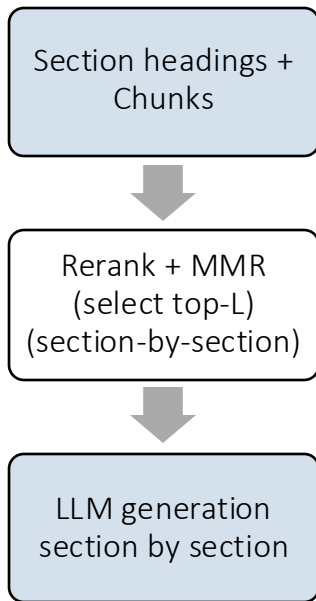
Selects the most relevant chunks while avoiding repetition.



## Method 2: Hierarchical RAG

(WikiGenBench: Plan-Retrieve-Read)

- Section structure is already planned.



```
SYSTEM_PROMPT = (
 "You are an expert encyclopedia writer. "
 "Write accurate, well-structured Wikipedia-style articles with citations."
)

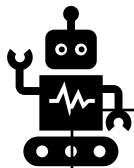
SECTION_PROMPT = """"\br/>I have a topic "{island_name}" and a section "{section}" that contains the following
documents:

{documents}

Based on the above information, write the "{section}" section of a Wikipedia article
\br/>about {island_name}.
You MUST cite the most relevant document for every sentence you write.
Use ONLY this exact citation format: "This is an example sentence.[1]" or "Another
sentence.[2][3]"
where the number inside the brackets refers to the Document number above.
Do NOT repeat the section title inside the content — just write the content
directly.
Do NOT write a "References", "See also", or "External links" section — only inline
citations like [1], [2], [3].
"""
```



## Method 3: Inter-Section Aware RAG



### InterSectionRAG

- Data Routing
- Blueprint Reception
- **Graph Evaluation**
- Execution Trigger

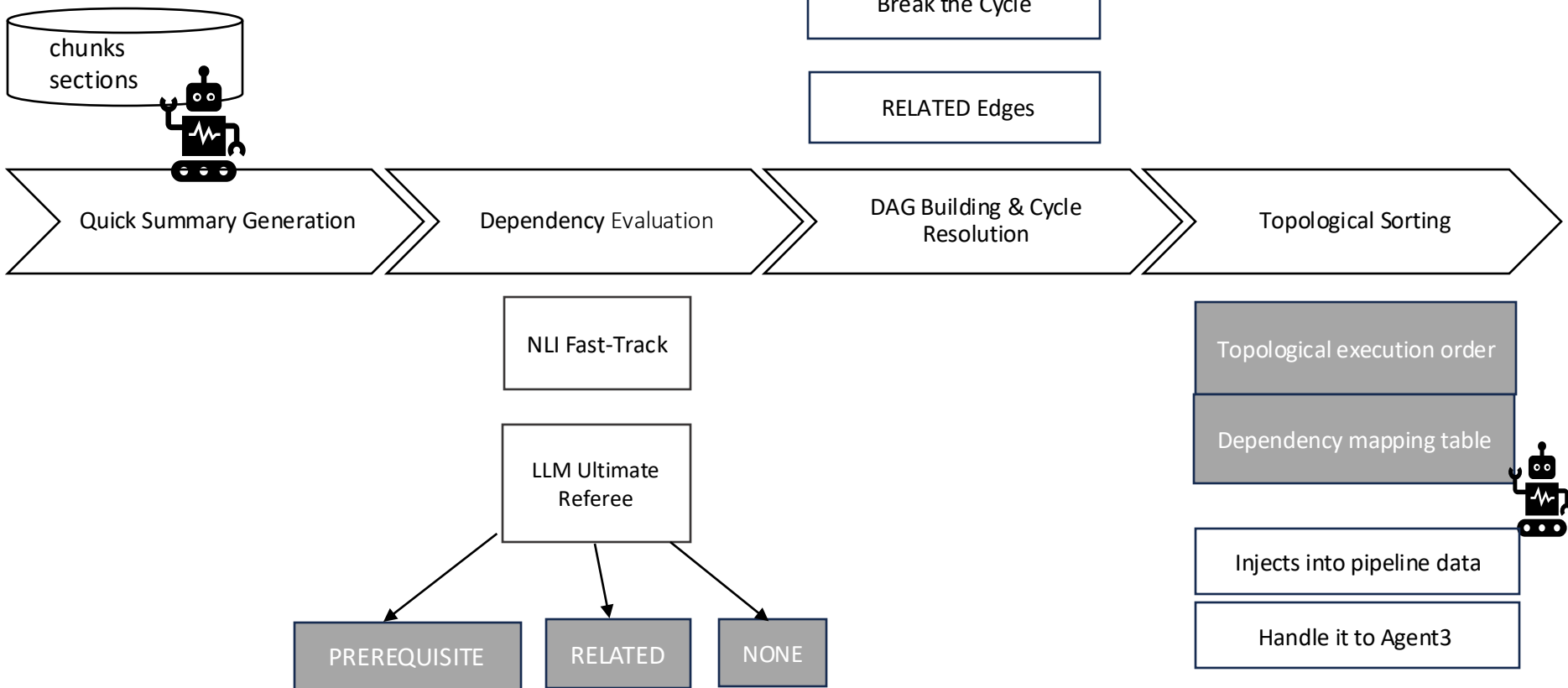
### Agent 2 (GraphAwareRAG)

- Quick Summary Extraction
- Dual-Gate Dependency Gatekeeper

### Agent 3 (Generator)

- Topological Execution Loop
- Dynamic Hard Context Injection
- Introspective Summarization

# Agent 2 Workflow



# Agent 2 – Important Prompts

## System Prompt

1

""You are a sterile, automated data-extraction API. Your output feeds directly into a programmatic pipeline. You possess ZERO conversational abilities. You do not greet, you do not explain, you do not apologize.

### CRITICAL RULES:

1. OUTPUT ONLY A SINGLE CONTINUOUS PARAGRAPH.
2. NO bullet points (1., 2., -, \*). NO line breaks.
3. ZERO CONVERSATIONAL FILLER. (BANNED PHRASES: "Based on the provided documents", "Here are the extracted facts", "I found", "However").
4. If there is no exact matching info for the Topic, output EXACTLY THIS STRING: "No relevant information regarding {section\_name} is available." ""

## User Prompt

1

""Topic: [{section\_name}]

Source Documents:  
{combined\_text}

=== EXPECTED OUTPUT EXAMPLES ===

Example for a successful match (Starts immediately with facts, dense text):  
Nishinoshima is a stratovolcano rising 3,000 meters from the ocean floor. It consists of basalt and andesite magmas, with the caldera almost filled in by 1973. The volcano provides a unique opportunity to study island-forming processes.

Example for NO match (Output this exact string):  
No relevant information regarding {section\_name} is available.  
=====

Output your final paragraph below: ""

## Prompt

2

""Topic A: [{sec\_a\_name}]  
Summary A: {sec\_a\_text}  
Topic B: [{sec\_b\_name}]  
Summary B: {sec\_b\_text}  
Question: Analyze the relationship between Topic A and Topic B. Choose EXACTLY ONE of the following three categories:  
1. "PREREQUISITE": Understanding Topic A is strictly required before learning Topic B (A determines B).  
2. "RELATED": They are interconnected and mutually reference each other, but there is no strict chronological or prerequisite order. They just shouldn't contradict each other.  
3. "NONE": They are practically independent.  
  
Output exactly and only one word: "PREREQUISITE", "RELATED", or "NONE". ""

Quick Summary  
Generation

1

Dependency Evaluation

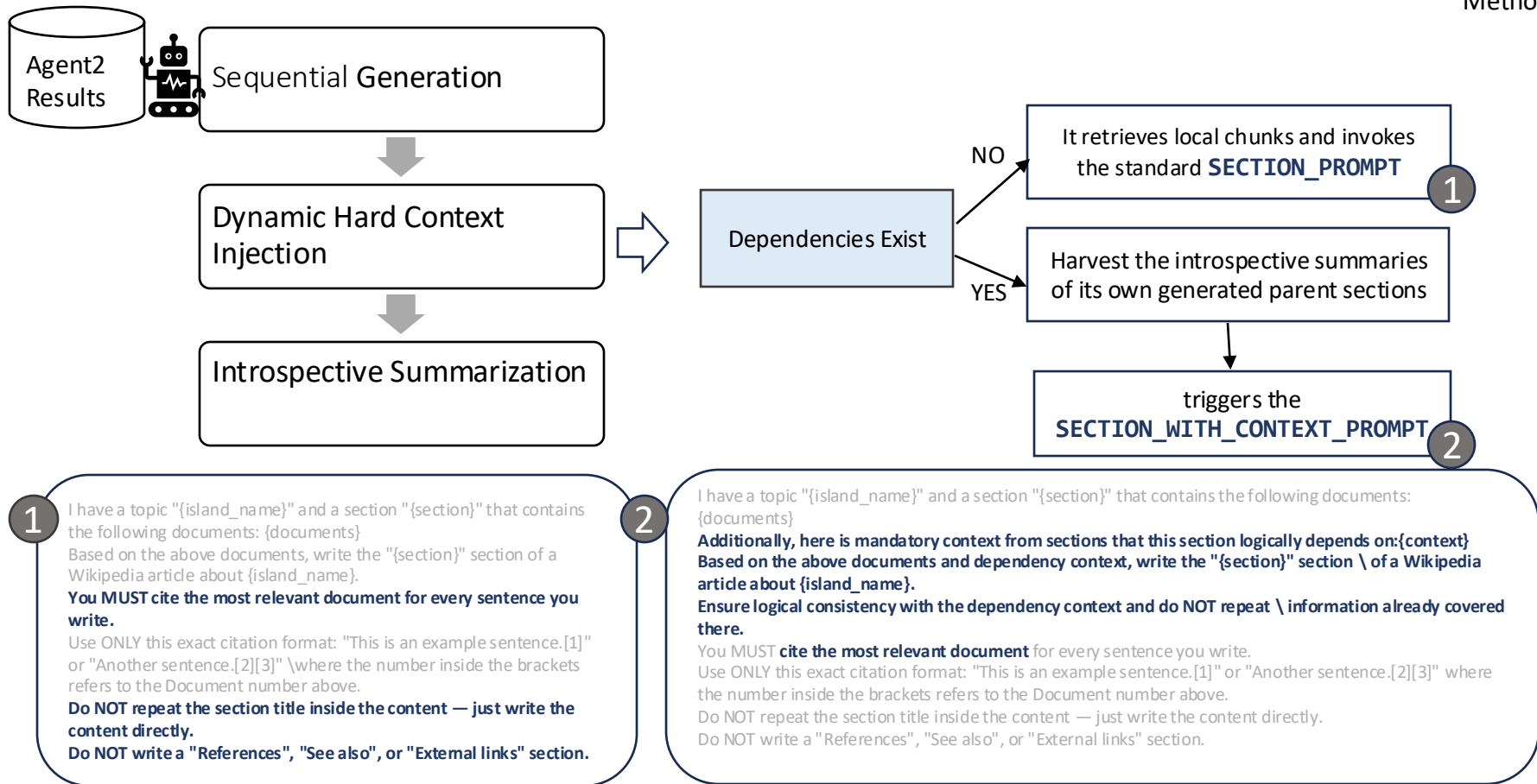
2

DAG Building & Cycle  
Resolution

Topological Sorting

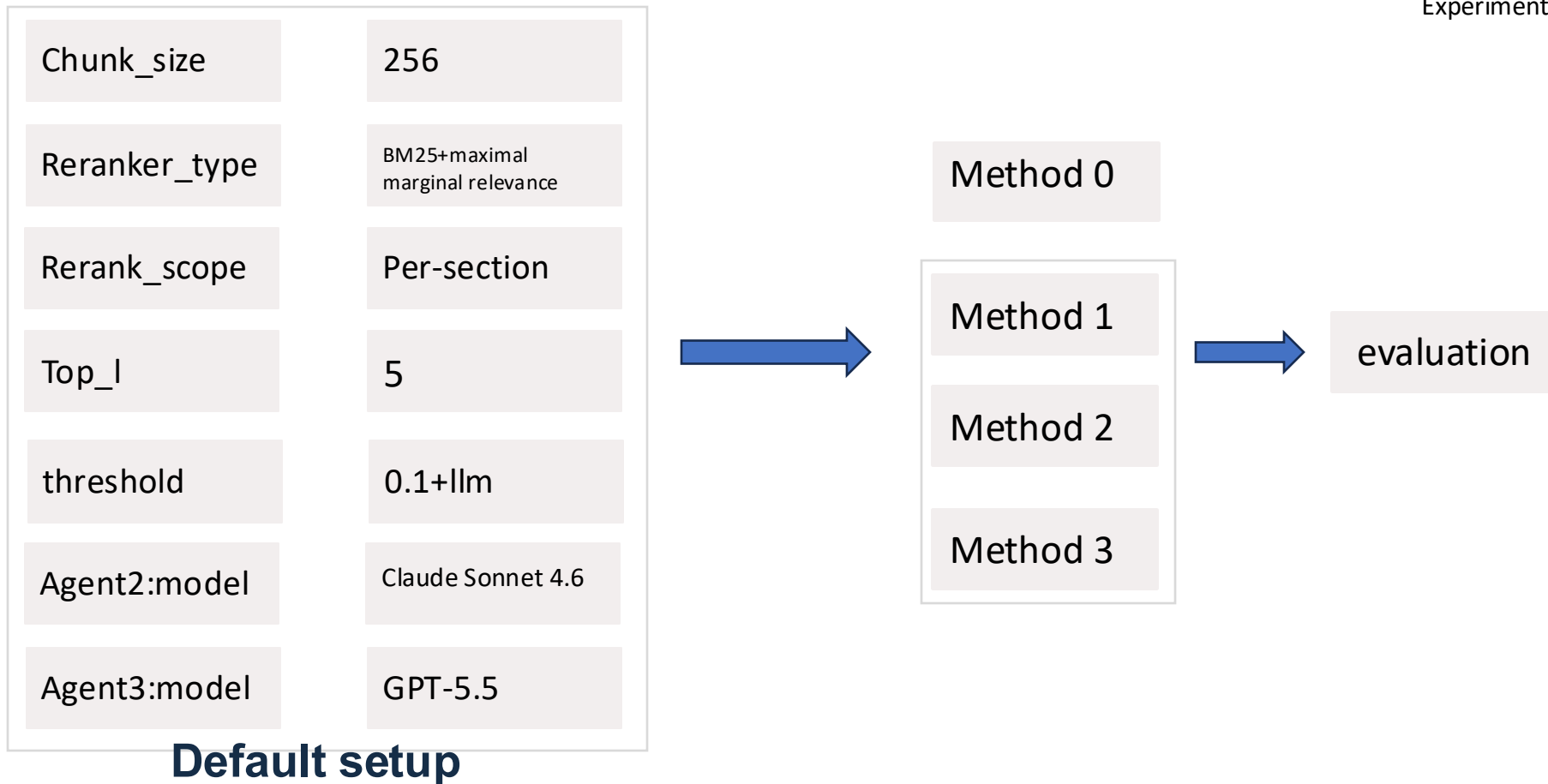


# Agent 3 Workflow





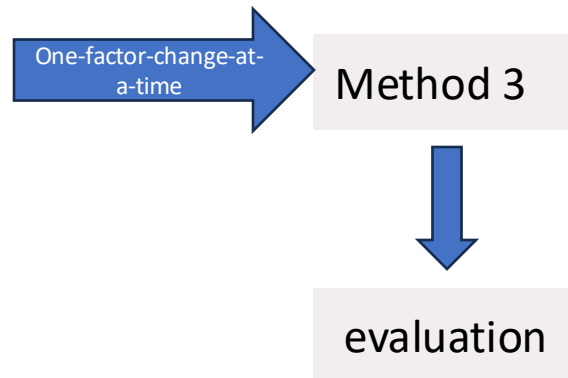
## Experimental setup (final article generation combinations)





## Experimental setup (final article generation combinations)

|                              |                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------|
| Chunk_size                   | 256                                                                                        |
| Reranker_type                | Spares:BM25+maximal marginal relevance<br>Dense:cross encoder + maximal marginal relevance |
| Rerank_scope                 | Per-section / global                                                                       |
| Top_l                        | 5                                                                                          |
| threshold                    | 0.1, 0.3, 0.6 + llm                                                                        |
| <a href="#">Agent2:model</a> | Claude Sonnet 4.6, Deepseek V3.2,Qwen 3.6 Plus,Gemini 3 Flash Preview                      |
| <a href="#">Agent3:model</a> | GPT-5.5, GPT-5.4, Claude Opus 4.7, Gemini 3.1 Pro Preview                                  |





# Evaluation Methodology

Generated Wikipedia-style island articles are evaluated across five complementary metric families. No single score fully captures encyclopedic quality — all families are interpreted jointly.

## Writing Quality

Readability, coherence,  
Wikipedia-style structure

## Lexical Informativeness

Lexical overlap with human-  
written reference

## Concept-Based Quality

Key concepts captured  
accurately and coherently

## Citation Verifiability

Factual claims cited and  
supported by evidence

## Cross-Sectional Consistency (CSCS)

Parent-section info preserved in  
child sections (Method 3 only)



# Evaluation Methodology

## Writing Quality

Assessed by an LLM-based evaluator. Each generated article is scored on four dimensions:

- Fluency
- Structure
- Organization
- Overall writing score

Evaluates readability, grammatical naturalness, and Wikipedia-style formatting on a 1–5 rubric scale.

## Lexical Informativeness

Measured with ROUGE-L and METEOR after removing citation markers from the generated text.

- ROUGE-L
- METEOR

These are reference-based metrics; scores reflect surface-level alignment, not factual correctness.



# Evaluation Methodology

## Concept-Based Quality

An LLM-based evaluator (Zheng et al., 2023) compares the generated article against the reference article and scores five dimensions:

- Concept Coverage
- Accuracy
- Relevance
- Organization
- Overall concept score

Assesses whether key reference concepts are captured accurately,  
Scored on a 1–5 rubric scale.

## Citation Verifiability

Each claim unit is matched with its citation markers and cited evidence. A claim is supported if the cited evidence entails it.

- Citation Rate
- Citation Precision
- Citation Recall
- Citation Link Precision

Scores on a 0–1 scale.



# Cross-Sectional Consistency Score (CSCS)

CSCS measures whether information stated in parent sections is preserved and consistent in dependent child sections

How it is computed

## 2. Count total dependency edges

```
{
 "Ecology": ["Geography"],
}
```

## 4. Get facts

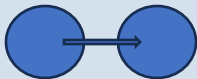
Fact 1: The island is volcanic.  
Fact 2: It has steep cliffs.  
Fact 3: heavy seasonal rainfall.

VS

Ecology text

Given Text A, does text A support text B?  
Using: `sentence-transformers/all-MiniLM-L6-v2`

## 1. Load the dependency graph



## 3. Extract facts form the parent section

Parent

Geography

The island is volcanic.  
It has steep cliffs. It  
receives heavy  
seasonal rainfall.

Child

Ecology

The island's volcanic  
soil supports sparse  
vegetation. Seasonal  
rainfall shapes plant  
growth.

## 5. NLI entailment + Similarity fallback

Fact 1: 0.9  
Fact 2: does not mention cliff: 0.1  
Fact 3: 0.9

$CSCS = \text{fact\_score\_sum} / \text{checked\_facts}$   
 $CSCS = 1.85 / 3$   
 $CSCS = 0.6167$

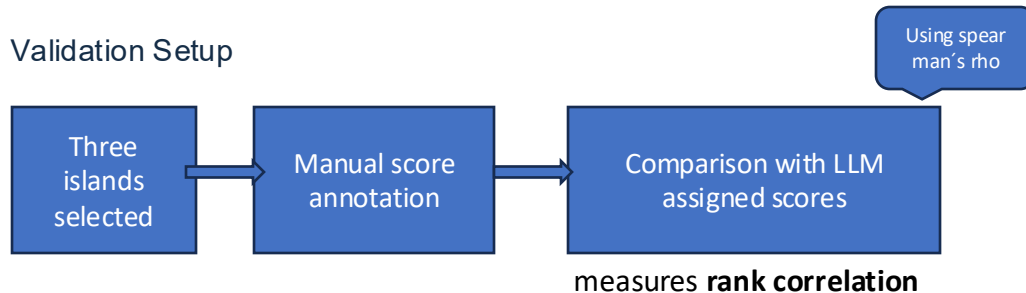


# LLM-as-Judge Validation

Because writing and concept scores rely on an LLM-based evaluator, a manual validation check was conducted to assess reliability.



## Validation Setup





# Error Analysis Methodology

## Error classes

| Error family               | Error class                | Detection logic                                                                   |
|----------------------------|----------------------------|-----------------------------------------------------------------------------------|
| Writing quality            | Writing error              | Low fluency, structure, organization, or overall writing score.                   |
| Content quality            | Content omission           | Low concept coverage, listed missing key concepts, or bottom-tail ROUGE-L/METEOR. |
| Content quality            | Content substitution       | Low concept accuracy or listed inaccurate/unsupported concepts.                   |
| Content quality            | Content addition           | Low concept relevance or off-topic generated content.                             |
| Content quality            | Content organization error | Low concept organization score.                                                   |
| Content quality            | Content quality error      | Low overall concept score.                                                        |
| Citation and verifiability | Citation coverage failure  | Low citation rate or citation recall.                                             |
| Citation and verifiability | Citation support failure   | Low citation precision or citation link precision.                                |
| Planning and graph         | Graph coherence failure    | Low CSCS; applicable only to method3.                                             |

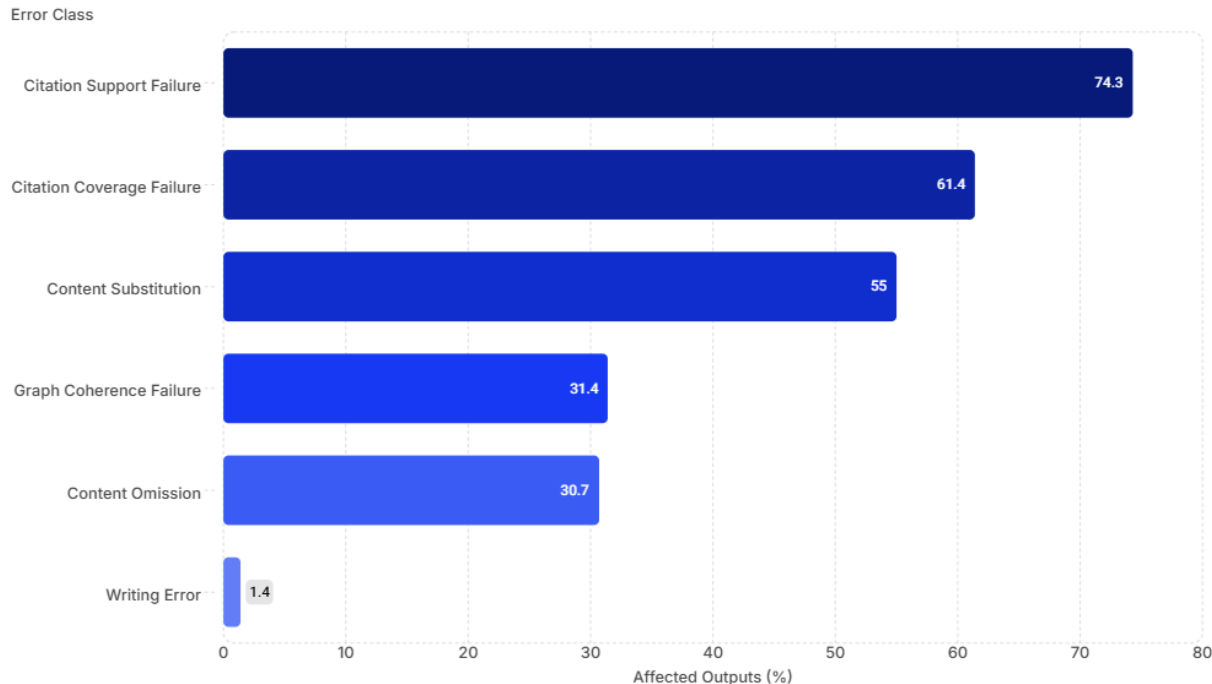
## Severity Thresholds

| Error source                                             | Medium severity      | High severity        |
|----------------------------------------------------------|----------------------|----------------------|
| Writing scores                                           | < 3.0                | < 2.5                |
| Concept coverage, relevance, organization, overall score | < 3.0                | < 2.5                |
| Concept accuracy                                         | < 3.5                | < 3.0                |
| Missing key concepts                                     | 1 item               | ≥ 2 items            |
| Inaccurate or unsupported concepts                       | –                    | Any listed item      |
| Citation rate                                            | < 0.80 or bottom 25% | < 0.50 or bottom 10% |
| Citation recall                                          | < 0.20 or bottom 25% | < 0.10 or bottom 10% |
| Citation precision                                       | < 0.25 or bottom 25% | < 0.10 or bottom 10% |
| Citation link precision                                  | < 0.10 or bottom 25% | < 0.05 or bottom 10% |
| CSCS                                                     | < 0.50 or bottom 25% | < 0.40 or bottom 10% |
| ROUGE-L / METEOR                                         | Bottom 25%           | Bottom 10%           |

Not using a standard 0.5 for every class



# Error Analysis: Dominant Failure Modes



The initial error analysis spans **140 generated outputs** across all default methods and Method 3 ablations. Citation-related failures dominate the error profile, while writing errors are rare — indicating that the primary weakness is factual grounding, not readability.



The unusually high initial citation failure rates (74.3%) motivated a dedicated debugging iteration of the citation-evaluation pipeline before final conclusions were drawn.



# Results: Default Method Comparison

Evaluated across 10 islands. Values are mean  $\pm$  standard deviation. **Bold** = best score per metric. Method 0 leads on writing and concept quality but produces zero citation grounding. After citation-evaluation debugging, Method 2 achieves the strongest citation scores among retrieval-based methods.

| Metric                                     | method0                              | method1                             | method2                              | method3            |
|--------------------------------------------|--------------------------------------|-------------------------------------|--------------------------------------|--------------------|
| <i>Writing quality</i>                     |                                      |                                     |                                      |                    |
| Fluency                                    | <b>4.700 <math>\pm</math> 0.483</b>  | 4.000 $\pm$ 0.000                   | 3.900 $\pm$ 0.316                    | 3.900 $\pm$ 0.316  |
| Structure                                  | <b>3.900 <math>\pm</math> 0.316</b>  | 3.800 $\pm$ 0.422                   | 3.100 $\pm$ 0.316                    | 3.400 $\pm$ 0.516  |
| Organization                               | <b>4.000 <math>\pm</math> 0.471</b>  | 3.900 $\pm$ 0.316                   | 3.300 $\pm$ 0.483                    | 3.500 $\pm$ 0.527  |
| Overall writing                            | <b>4.199 <math>\pm</math> 0.320</b>  | 3.901 $\pm$ 0.159                   | 3.433 $\pm$ 0.227                    | 3.600 $\pm$ 0.345  |
| <i>Concept quality</i>                     |                                      |                                     |                                      |                    |
| Coverage                                   | <b>3.800 <math>\pm</math> 0.422</b>  | 3.200 $\pm$ 0.422                   | 3.200 $\pm$ 0.633                    | 3.300 $\pm$ 0.483  |
| Accuracy                                   | <b>3.800 <math>\pm</math> 0.633</b>  | 3.200 $\pm$ 0.422                   | 3.200 $\pm$ 0.633                    | 3.600 $\pm$ 0.516  |
| Relevance                                  | <b>4.500 <math>\pm</math> 0.527</b>  | 4.400 $\pm$ 0.516                   | 4.300 $\pm$ 0.483                    | 4.400 $\pm$ 0.516  |
| Organization                               | <b>4.100 <math>\pm</math> 0.316</b>  | <b>4.100 <math>\pm</math> 0.316</b> | 3.600 $\pm$ 0.699                    | 3.500 $\pm$ 0.527  |
| Overall concept                            | <b>4.050 <math>\pm</math> 0.350</b>  | 3.725 $\pm$ 0.362                   | 3.575 $\pm$ 0.514                    | 3.700 $\pm$ 0.453  |
| <i>Lexical informativeness</i>             |                                      |                                     |                                      |                    |
| ROUGE-L                                    | 15.792 $\pm$ 2.362                   | 13.860 $\pm$ 3.373                  | <b>16.257 <math>\pm</math> 2.596</b> | 15.699 $\pm$ 2.542 |
| METEOR                                     | <b>26.316 <math>\pm</math> 4.117</b> | 16.099 $\pm$ 9.173                  | 22.148 $\pm$ 8.020                   | 22.617 $\pm$ 9.270 |
| <i>Citation grounding before debugging</i> |                                      |                                     |                                      |                    |
| Citation recall                            | 0.000 $\pm$ 0.000                    | <b>0.231 <math>\pm</math> 0.205</b> | 0.180 $\pm$ 0.138                    | 0.202 $\pm$ 0.192  |
| Citation precision                         | 0.000 $\pm$ 0.000                    | <b>0.233 <math>\pm</math> 0.204</b> | 0.194 $\pm$ 0.160                    | 0.203 $\pm$ 0.191  |
| Citation link prec.                        | 0.000 $\pm$ 0.000                    | <b>0.105 <math>\pm</math> 0.117</b> | 0.070 $\pm$ 0.067                    | 0.061 $\pm$ 0.068  |
| Citation rate                              | 0.000 $\pm$ 0.000                    | <b>0.979 <math>\pm</math> 0.045</b> | 0.941 $\pm$ 0.101                    | 0.975 $\pm$ 0.054  |
| <i>Citation grounding after debugging</i>  |                                      |                                     |                                      |                    |
| Citation recall                            | 0.000 $\pm$ 0.000                    | 0.231 $\pm$ 0.205                   | <b>0.464 <math>\pm</math> 0.150</b>  | 0.330 $\pm$ 0.165  |
| Citation precision                         | 0.000 $\pm$ 0.000                    | 0.233 $\pm$ 0.204                   | <b>0.481 <math>\pm</math> 0.144</b>  | 0.345 $\pm$ 0.168  |
| Citation link prec.                        | 0.000 $\pm$ 0.000                    | 0.105 $\pm$ 0.117                   | <b>0.178 <math>\pm</math> 0.076</b>  | 0.148 $\pm$ 0.088  |
| Citation rate                              | 0.000 $\pm$ 0.000                    | <b>0.979 <math>\pm</math> 0.045</b> | 0.962 $\pm$ 0.068                    | 0.957 $\pm$ 0.074  |
| <i>Dependency consistency</i>              |                                      |                                     |                                      |                    |
| CSCS                                       | –                                    | –                                   | –                                    | 0.507 $\pm$ 0.076  |



Iterative Debugging



# Ablation study on Method 3 components

Each experiment varies one component or parameter of Method 3 in isolation — planning model, generation model, retrieval threshold, reranking scope, and reranker type. Results indicate **metric-specific improvement potential**, not a single optimized configuration.

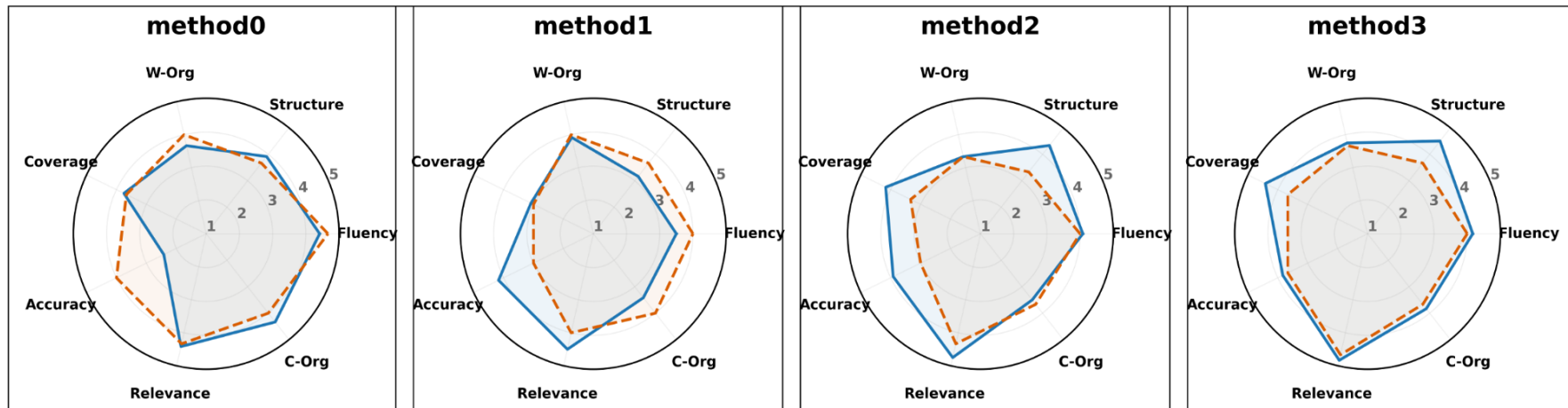
Table 7: Method 3 component and parameter experiments.

| Metric                  | Default Method 3   | Best Setting                  | Best Result                          | Gain   |
|-------------------------|--------------------|-------------------------------|--------------------------------------|--------|
| Writing                 | $3.600 \pm 0.345$  | Threshold 0.6                 | <b><math>3.699 \pm 0.368</math></b>  | +0.099 |
| ROUGE-L                 | $15.699 \pm 2.542$ | Agent 3: Claude Opus 4.7      | <b><math>17.182 \pm 2.914</math></b> | +1.483 |
| METEOR                  | $22.617 \pm 9.270$ | Agent 3: Claude Opus 4.7      | <b><math>28.096 \pm 6.570</math></b> | +5.479 |
| Concept                 | $3.700 \pm 0.453$  | Agent 2: Qwen3.6 Plus         | <b><math>3.800 \pm 0.387</math></b>  | +0.100 |
| Citation recall         | $0.202 \pm 0.192$  | Cross-encoder + MMR reranking | <b><math>0.344 \pm 0.199</math></b>  | +0.142 |
| Citation precision      | $0.203 \pm 0.191$  | Cross-encoder + MMR reranking | <b><math>0.362 \pm 0.204</math></b>  | +0.159 |
| Citation link precision | $0.061 \pm 0.068$  | Cross-encoder + MMR reranking | <b><math>0.183 \pm 0.135</math></b>  | +0.122 |
| CSCS                    | $0.507 \pm 0.076$  | Agent 3: GPT-5.4              | <b><math>0.533 \pm 0.068</math></b>  | +0.026 |



Result

# Human vs. LLM Evaluator Agreement



— Human    - - - LLM-agent

| Measure          | Unit of comparison                        | $n$ | Spearman's $\rho$ | $p$   |
|------------------|-------------------------------------------|-----|-------------------|-------|
| Rank correlation | Island $\times$ method $\times$ dimension | 84  | 0.279             | 0.010 |



## Discussion & Key Findings

Inter-section awareness is promising but does not consistently outperform hierarchical RAG. The main hypothesis is only partially supported: dependency-aware context can help structure long-form generation, but does not yet produce clear overall gains.

Method 0 Leads on Fluency but Cannot Be Grounded

Method 2 Achieves the Strongest Citation Grounding

After citation-evaluation debugging, hierarchical PRR leads on citation recall (0.464), precision (0.481), and link precision (0.178).

Method 3 Is Competitive — Evidence Selection Is the Bottleneck

Inter-section aware generation does not consistently surpass Method 2 in the default configuration.



## Limitations

### Generalizability

Findings are based on 10 islands

### Evaluation Constraints

Method 3 dependency graphs are constructed automatically. Human validation covers only 3 islands. CSCS and citation diagnostics are limited to Method 3 outputs.

## Future Work



### Broader Evaluation

Evaluate on more diverse topics beyond islands to test generalizability across domains and article types.

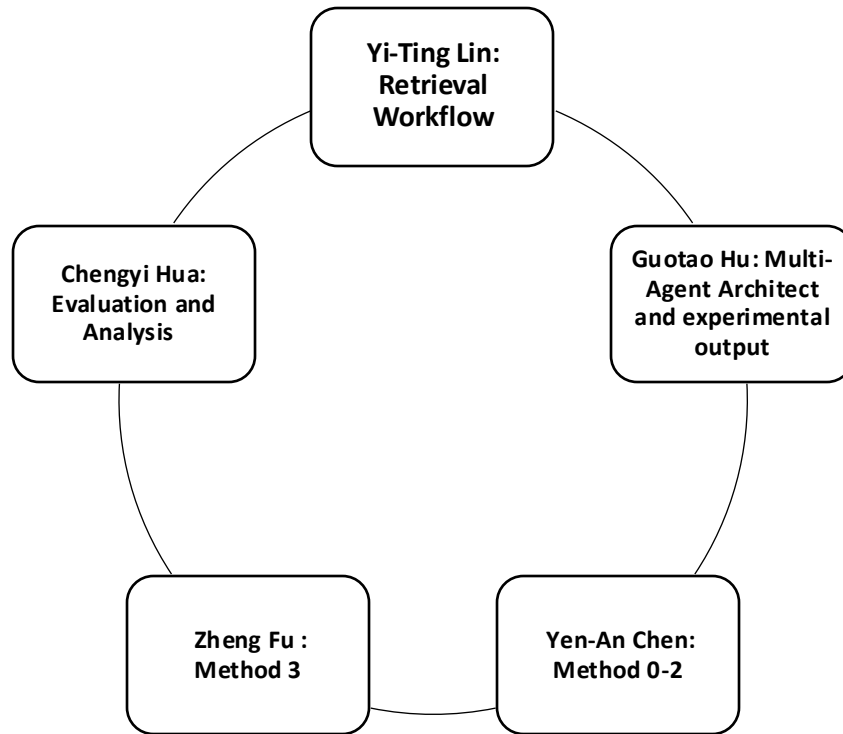
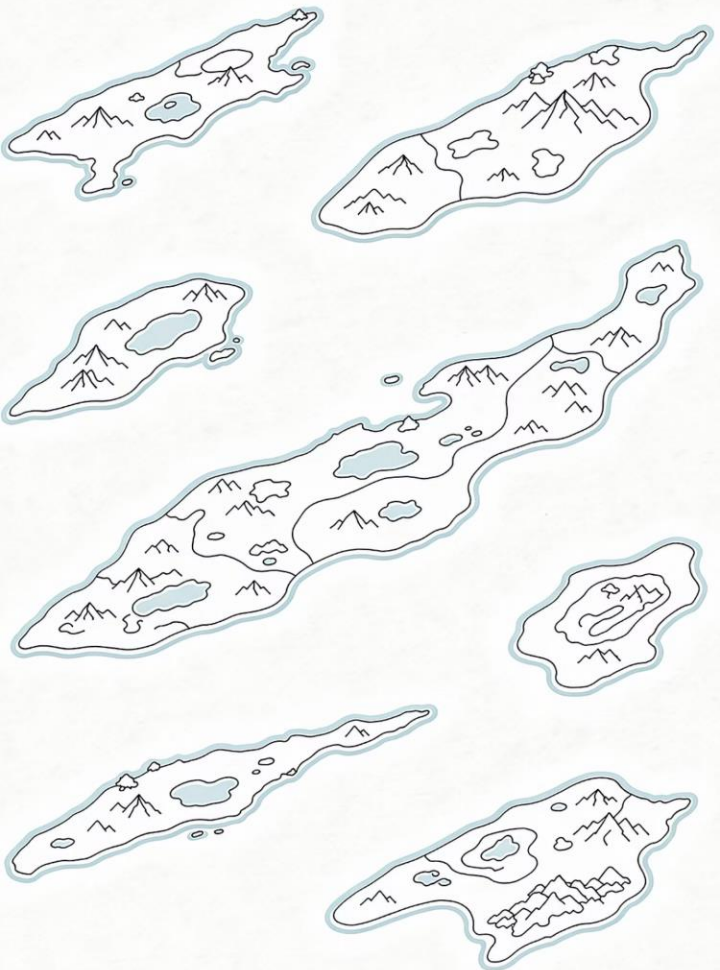


### Better Dependency Graphs

Improve Method 3's dependency graph construction to reduce repetition and ensure sub-sections provide genuinely incremental information.



## Team – Who did what



# Appendix



# Citation Metrics — Formal Definitions

The four citation verifiability metrics are defined as follows:

|                                                                           |                                                                                   |                                                                                       |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Citation Rate</b><br># cited sentences / # total sentences             |  | <b>Citation Precision</b><br># supported cited sentences / # cited sentences          |
| <b>Citation Recall</b><br># supported cited sentences / # total sentences |  | <b>Citation Link Precision</b><br># supported citation links / # total citation links |

A sentence is 'supported' if the cited evidence entails the claim. All scores are on a 0–1 scale.



# Writing Quality & Lexical Informativeness

LLM-based and reference-based metrics for surface-level article quality.

## Writing Quality — LLM Judge (1–5 scale)

An LLM evaluator scores each generated article using the same rubric as WIKIGENBENCH. Higher = better.

### Fluency

Is the text grammatically natural and easy to read?

### Structure

Does the article follow Wikipedia-style section organization?

### Organization

Is information logically sequenced and coherent?

### Overall Writing

Holistic writing quality — aggregates the three dimensions above

## Concept-Based Quality — LLM Judge (1–5 scale)

An LLM compares the generated article against the reference and scores five dimensions. Higher = better.

### Coverage

Are the key reference concepts present in the generated article?

### Accuracy

Are the concepts factually correct and not distorted?

### Relevance

Is the content on-topic and free of hallucinated additions?

### Organization

Are concepts presented in a logical, coherent order?

### Overall Concept

Holistic concept quality — aggregates the four dimensions above



# Writing Quality & Lexical Informativeness Prompt

LLM-based and reference-based metrics for surface-level article quality.

```
AGENT4_WRITING_SYSTEM_PROMPT = ""
You are Agent 4, an expert evaluator of Wikipedia-style article writing.

You evaluate ONLY writing quality.
Do not evaluate factual correctness.
Do not evaluate citation correctness.
Do not compare the article against a reference article.
Do not reward the article simply for being long.
Do not punish the article for missing facts unless that affects writing structure or organization.

Do not explain step by step.
Do not include reasoning.
Return only the final JSON object.
Return only valid JSON.
"".strip()
```

```
AGENT4_WRITING_PROMPT = ""
Evaluate the writing quality of the following generated Wikipedia-style article.

Target entity: {island_name}
Generation method: {method_name}

Rate the article on exactly three metrics, each from 0 to 5.

1. fluency_score
Evaluate grammar, readability, sentence-level clarity, and whether the prose sounds natural.

0 = unreadable or incoherent
1 = very poor fluency with major grammar/readability issues
2 = weak fluency; understandable but awkward or repetitive
3 = acceptable fluency with some issues
4 = good fluency; mostly clear and readable
5 = excellent fluency; polished Wikipedia-style prose

2. structure_score
Evaluate the article's section structure. This corresponds to section/outline quality.
Check whether the article has appropriate section headings, whether sections are meaningful,
and whether the structure fits a Wikipedia-style article about the target entity.

0 = no useful structure
1 = very poor section structure
2 = weak structure; headings are missing, confusing, or poorly chosen
3 = acceptable structure but incomplete or uneven
4 = good section structure with mostly appropriate headings
5 = excellent Wikipedia-style structure with clear and comprehensive headings

3. organization_score
Evaluate logical organization and coherence within and across sections.
Check whether ideas flow in a sensible order and whether the article avoids unnecessary repetition.

0 = disorganized or incoherent
1 = very poor organization
2 = weak organization with major flow or repetition problems
3 = acceptable organization with some issues
4 = good organization and mostly logical flow
5 = excellent organization; coherent, well connected, and easy to follow

Return ONLY valid JSON in this exact format:

{{
 "fluency_score": 0,
 "structure_score": 0,
 "organization_score": 0,
 "brief_rationale": "one concise sentence explaining the scores"
}}
```



## APPENDIX A

# Writing Quality & Lexical Informativeness Prompt

LLM-based and reference-based metrics for surface-level article quality.

```
AGENT4_CONCEPT_SYSTEM_PROMPT = """
You are Agent 4, an expert evaluator of Wikipedia-style article content.

You evaluate concept-level content quality by comparing a generated article
against a human-written reference Wikipedia article.

Focus on key concepts, important facts, topical coverage, and conceptual accuracy.
Do not evaluate citation formatting.
Do not evaluate prose fluency unless it affects conceptual organization.
Do not require exact wording from the reference article.
Do not reward unnecessary length.
Do not punish different but reasonable section names if the key concepts are covered.

Do not explain step by step.
Do not include reasoning.
Return only the final JSON object.
Return only valid JSON.
""".strip()
```

```
AGENT4_CONCEPT_PROMPT = """
Evaluate the concept-level quality of the generated Wikipedia-style article by comparing it
against the human-written reference article.

Target entity: {island_name}
Generation method: {method_name}

You must rate exactly four metrics, each from 0 to 5.

1. concept_coverage_score
How well does the generated article cover the key concepts from the reference article?

0 = covers almost none of the key concepts
1 = covers very few key concepts
2 = covers some key concepts but misses many important ones
3 = covers several important concepts but is incomplete
4 = covers most key concepts
5 = covers nearly all central concepts from the reference article

2. concept_accuracy_score
Are the concepts that appear in the generated article accurate with respect to the reference?

0 = mostly inaccurate or misleading
1 = many serious inaccuracies
2 = several inaccuracies or distortions
3 = mostly accurate with some issues
4 = accurate with only minor issues
5 = highly accurate representation of the reference concepts

3. concept_relevance_score
Is the generated article focused on the target entity and free from irrelevant or off-topic content?

0 = mostly irrelevant
1 = much irrelevant content
2 = some relevant content but substantial off-topic material
3 = mostly relevant with some off-topic or weakly related material
4 = relevant and focused
5 = highly focused on the target entity

4. concept_organization_score
Are the covered concepts arranged in a sensible Wikipedia-style conceptual structure?

0 = concepts are incoherent or randomly arranged
1 = very poor conceptual organization
2 = weak organization with major placement issues
3 = acceptable organization
4 = good conceptual organization
5 = excellent conceptual organization

Return ONLY valid JSON in this exact format:

{
 "concept_coverage_score": 0,
 "concept_accuracy_score": 0,
 "concept_relevance_score": 0,
 "concept_organization_score": 0,
}
```



## Appendix

### APPENDIX A

#### Citation Verifiability — 4 Metrics (0–1 scale)

A sentence is 'supported' if the cited evidence entails the claim.

##### Citation Rate

$\# \text{ cited sentences} \div \# \text{ total sentences}$

→ What share of sentences include any citation at all?

##### Citation Precision

$\# \text{ supported cited sentences} \div \# \text{ cited sentences}$

→ Of cited sentences, how many are actually backed up by evidence?

##### Citation Recall

$\# \text{ supported cited sentences} \div \# \text{ total sentences}$

→ Of all sentences, how many are verifiably supported?

##### Citation Link Precision

$\# \text{ supported citation links} \div \# \text{ total citation links}$

→ Of all individual links, how many point to genuinely supporting evidence?

##### Current status

The island remains uninhabited, with empty high-rise buildings, seawalls and other mine-era st spray.[1] Contemporary travel writing and tourism media commonly present Hashima as an aba destination associated with heritage tourism, photography and urban exploration.[2][3] Its rui academic and creative work, including the AHRC-linked project "The Future of Ruins: Reclaim Hashima Island" and the related 2013 performance "Return to Battleship Island".[4]

| Metric                  | Unit being judged        | What counts?                                                                                                                | What it penalizes                                                                                               |
|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Citation Rate           | Sentence                 | A sentence counts if it contains at least one citation marker. It does not check whether the citation supports the claim.   | Uncited sentences.                                                                                              |
| Citation Precision      | Cited sentence           | A cited sentence counts as supported if at least one of its citation links supports the sentence or a claim unit inside it. | Cited sentences where none of the citations support the claim.                                                  |
| Citation Recall         | All sentences            | A sentence counts if it is verifiably supported by at least one citation.                                                   | Any sentence that is not verifiably supported, including uncited sentences and cited-but-unsupported sentences. |
| Citation Link Precision | Individual citation link | Each citation link is checked separately; a link counts only if that specific source supports the sentence or claim unit.   | Extra, irrelevant, weak, or decorative citations, even if the sentence has one good citation.                   |



## Lexical Informativeness — Reference-Based

Computed against the human-written reference article after removing citation markers. Not a direct measure of factual correctness.

### ROUGE-L

Longest common subsequence overlap with the reference.

→ Captures how much of the reference's phrasing and structure appears in the generated text.

### METEOR

Word-level similarity with stemming and synonymy matching.

→ More lenient than ROUGE-L; rewards paraphrasing and near-matches.

Both metrics measure lexical similarity to a reference — they do not verify factual accuracy.



Initial error frequency across all 140 outputs before citation-evaluation debug.

| Error Class                | Affected Outputs  | High-Severity Outputs |
|----------------------------|-------------------|-----------------------|
| Citation support failure   | 104 / 140 (74.3%) | 75 / 140 (53.6%)      |
| Citation coverage failure  | 86 / 140 (61.4%)  | 58 / 140 (41.4%)      |
| Content substitution       | 77 / 140 (55.0%)  | 6 / 140 (4.3%)        |
| Graph coherence failure    | 44 / 140 (31.4%)  | 12 / 140 (8.6%)       |
| Content omission           | 43 / 140 (30.7%)  | 19 / 140 (13.6%)      |
| Writing error              | 2 / 140 (1.4%)    | 2 / 140 (1.4%)        |
| Content quality error      | 2 / 140 (1.4%)    | 0 / 140 (0.0%)        |
| Content organization error | 1 / 140 (0.7%)    | 1 / 140 (0.7%)        |
| Content addition           | 0 / 140 (0.0%)    | 0 / 140 (0.0%)        |

Default-method error rates after debug. High-severity rates in parentheses.

| Error Class               | method0     | method1   | method2   | method3   |
|---------------------------|-------------|-----------|-----------|-----------|
| Content omission          | 20% (10%)   | 70% (40%) | 30% (10%) | 40% (30%) |
| Content substitution      | 10% (10%)   | 80% (0%)  | 40% (0%)  | 50% (0%)  |
| Citation coverage failure | 100% (100%) | 60% (20%) | 30% (10%) | 50% (30%) |
| Citation support failure  | 100% (100%) | 70% (30%) | 20% (10%) | 40% (20%) |
| Graph coherence failure   | –           | –         | –         | 40% (20%) |



APPENDIX A

# Cross-Sectional Consistency Score (CSCS)

The only metric that directly measures cross-section coherence — exclusive to Method 3.

## What Does CSCS Measure?

CSCS checks whether facts stated in a parent section are preserved and consistent in dependent child sections — capturing the coherence benefit of Method 3's dependency graph.

### How It Works

01

#### Extract

Up to 5 fact-like units are extracted from each parent section per dependency edge in the Agent 2 graph.

02

#### Compare

Each fact is compared to the corresponding child section using entailment. If entailed:  $s_i = 1$ . Otherwise: semantic-similarity fallback score  $s_i \in [0, 1]$ .

03

#### Aggregate

$CSCS = (1/N) \times \sum s_i$  — the mean support score over all  $N$  checked dependency facts.

### Formula Breakdown

$$CSCS = (1/N) \times \sum s_i$$

The final score is the average support across all parent-child fact comparisons.

$N$

Total number of dependency facts checked across all parent-child edges in the article.

$$s_i \in [0, 1]$$

Support score for the  $i$ -th fact.  $s_i = 1$  if the child section entails the parent fact; otherwise a semantic-similarity score is used as fallback.



# The Island Dataset

Ten geographically diverse islands were selected to maximise inter-disciplinary dependency richness — cases where volcanic geology, resulting ecology, and human history are tightly coupled and measurable.



## Volcanically Active

**Surtsey** (emerged 1963), **Nishinoshima**, and **Hunga Tonga** (catastrophic 2022 eruption) represent cases where Wikipedia entries are severely outdated, testing the system's ability to incorporate recent evidence.



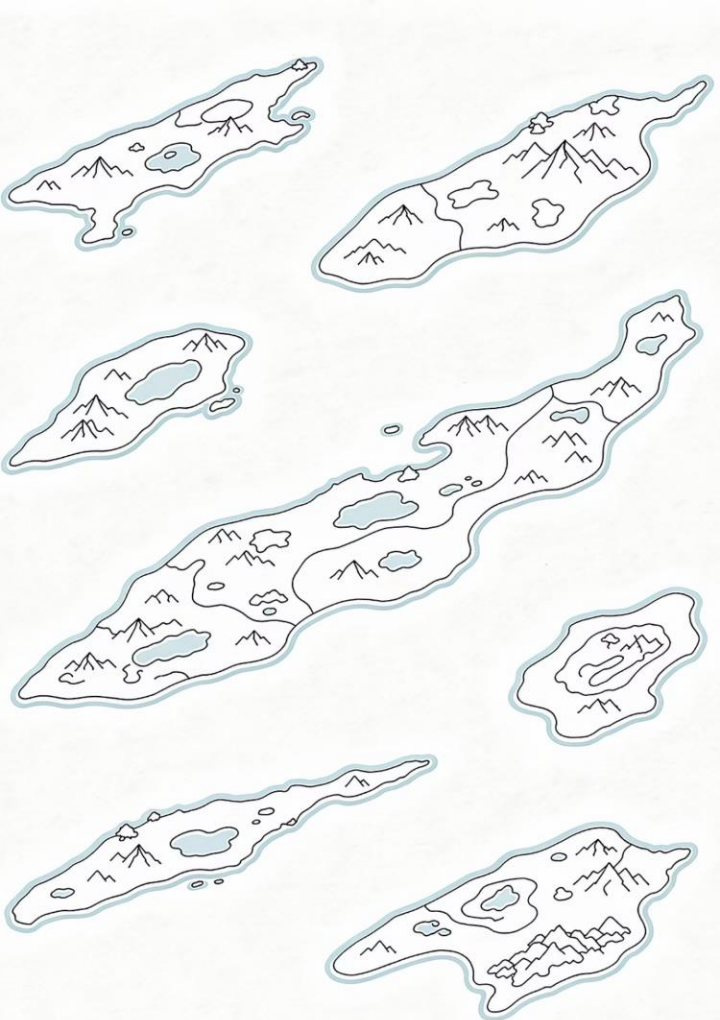
## Uninhabited & Protected

**Jarvis Island** and **Howland Island** are U.S. unincorporated territories in the Pacific — ecologically significant but data-sparse, testing retrieval under low-resource conditions.



## Human History Focus

**Hashima Island** (Japan's infamous abandoned coal-mining city) and **Jethou** (Channel Islands) feature rich human and administrative histories, where concept accuracy and narrative coherence are critical.



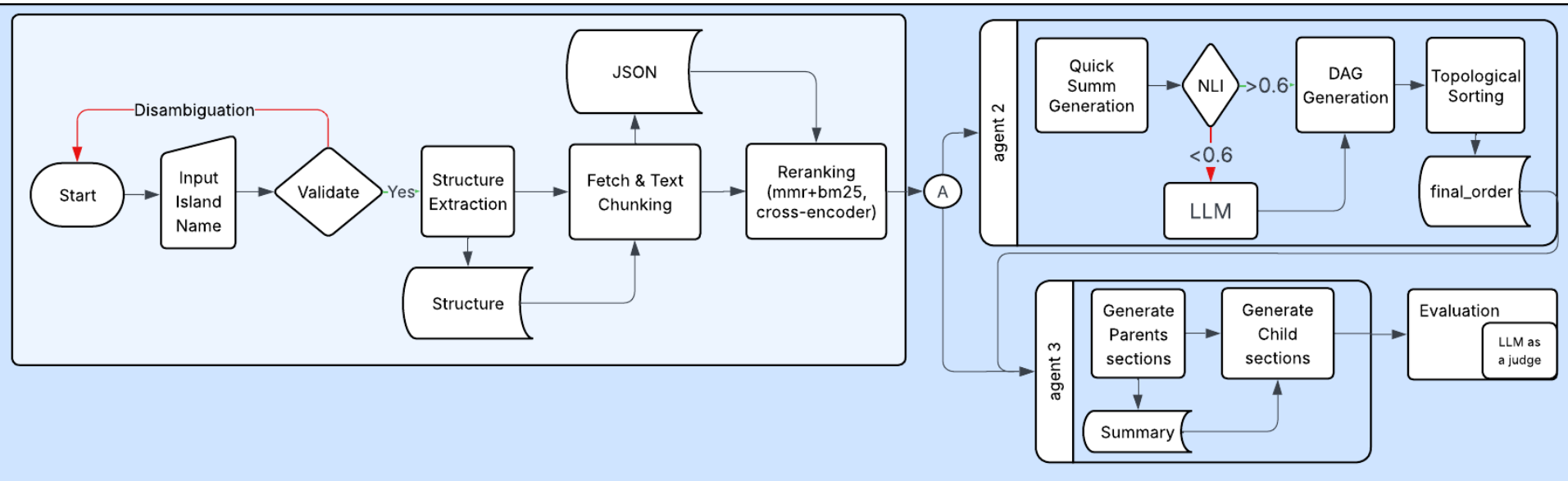
# Maximal Marginal Relevance (MMR)

Carbonell & Goldstein (1998)

$$\text{MMR} \stackrel{\text{def}}{=} \arg \max_{D_i \in R \setminus S} \left[ \lambda \text{Sim}_1(D_i, Q) - (1 - \lambda) \max_{D_j \in S} \text{Sim}_2(D_i, D_j) \right]$$

- Q: query (island name, or island name + section heading)
- R: all candidate chunks
- S: already selected chunks
- $D_i$  : candidate chunk being evaluated
- $D_j$  : already selected chunk
- Sim1: similarity between chunk and query
- Sim2 : similarity between chunks
- $\lambda=0.5$ : balance between relevance and diversity (0: diversity only, 1: relevance only)

# Flowchart



# Method3 - Lower Redundancy (4 times UNESCO in M2)

## History

Hashima Island developed from a small coal-mining and fishing site into a major undersea coal-mining settlement off Nagasaki.[1][3]

Coal had long been known in the surrounding area, with people on the island using it as household fuel before industrial mining expanded.[2] During Japan's Meiji period, the island shifted toward coal-power

Mitsubishi transformed the island into a concentrating mining site, with thousands of tons of coal annually shipped from an apartment block on the island. The building of significant structures were built to support the

Hashima's dense built-up area made it one of the most densely populated islands in the world, with 83,500 people per square kilometer more than 5,000 residents.

The island's history also includes its role in the Japanese colonial status, and the postwar community coexist with these earlier histories of coercion, making it a site of historical significance.

Hashima's prosperity declined as Japan's energy economy shifted away from coal. The mine closed in 1974, and the island was abandoned soon afterwards. In the decades after abandonment, Hashima's concrete buildings deteriorated under exposure to wind, and fire, turning the former mining town into a prominent urban ruin.[1][5] The island later became a major tourist destination and gained attention as both a tourist destination and a symbol of Japan's industrial past.[1][5]

Hashima forms part of the UNESCO-listed "Sites of Japan's Meiji Industrial Revolution: Iron and Steel, Shipbuilding and Coal Mining," but its listing and interpretation have generated diplomatic tension over how to represent wartime labor history.[1][3]

## World Heritage Site approval controversy

Hashima Island was included among the 23 components of Japan's "Sites of Japan's Meiji Industrial Revolution," a World Heritage nomination intended to represent Japan's rapid transformation into an industrial power in the late 19th century.[2] UNESCO's World Heritage Committee designated the group in 2015, and Hashima became part of the World Heritage site despite objections centered on wartime labor history.[3][5] South Korea objected that seven of the nominated industrial sites, including Hashima, had used forced labor during World War II and argued that those sites should be removed from the bid.[1] Korean accounts cited Hashima as

the most notorious of the sites, believing that many workers died from exposure to the elements, imperial and wartime history, and that the island's history was often interpreted in a way that acknowledged Japanese imperialism rather than Japan's industrialization rather than Japan's Industrial Heritage. The island's history was often interpreted in a way that acknowledged Japanese imperialism rather than Japan's industrialization rather than Japan's Industrial Heritage. The island's history was often interpreted in a way that acknowledged Japanese imperialism rather than Japan's industrialization rather than Japan's Industrial Heritage.

Hashima's World Heritage status was a result of Japan's industrialization rather than Japan's Industrial Heritage. The island's history was often interpreted in a way that acknowledged Japanese imperialism rather than Japan's industrialization rather than Japan's Industrial Heritage.

## In popular culture

Hashima Island has been widely represented through fictional and mediated images, with James Bond fans recognizing it as the location of Silva's lair in the 2012 film \*Skyfall\*.[2][4] The island has also appeared in National Geographic's \*One Strange Town\* coverage emphasizing its abandoned concrete landscape and decay.[4] Survival-horror games and immersive media have also evoked Hashima, including \*The Closed Ward\* and later virtual-reality projects, museum installations, and commercial walking simulators.[2] Tourism campaigns and the island's 2015 UNESCO World Heritage listing have popularized images of its ruins as picturesque decay, helping make Gunkanjima a major tourist attraction in Nagasaki.[2][3] However, critics argue that such popular representations can shape collective memory of Hashima and affect debates over Japan's wartime forced labor and other abuses associated with the island.[1] Critical studies of \*Skyfall\* and related media suggest that entertainment-focused portrayals may reinforce a triumphalist view of Japan's industrial heritage while overlooking the trauma, loss, and reparation.[1]

# Method3 - Lower Redundancy (2 times UNESCO in M3)

## World Heritage Site approval controversy

Hashima Island was included in Japan's 23-site nomination for the UNESCO listing "Sites of Japan's Meiji Industrial Revolution", which presented the sites as evidence of Japan's rapid industrialization

South Korea objected to the nomination on the grounds that seven of the sites were destroyed during the Second World War, and argued that those sites should be excluded from the nomination

Hashima was among the most contentious locations, with Korean critics arguing that the island was a site of mines and that many died from exposure or overwork.[1]

Japanese proponents of the nomination argued that the World Heritage status would promote the island's wartime mobilization from the Korean Peninsula, while noting that Hashima was acquired by Mitsubishi in 1890.[3]

UNESCO approved the Meiji industrial sites, including Hashima, as World Heritage Sites in 2015, despite South Korea's objections over how the sites' wartime labor history would be represented.

Some critics argued that World Heritage status itself was not necessarily a celebration of the island's place for historical self-examination rather than solely as a celebration of Japanese industrialization.

The dispute continued after approval, especially after the Industrial Heritage Committee in Paris. Korean Foreign Minister Kang Kyung-wha told UNESCO that its explanation of the sites' history was insufficient and asked UNESCO to consider removing Hashima's World Heritage status.

Supporters of the Japanese interpretation, including the Industrial Heritage Centre, had gathered extensive primary materials and testimonies from former Hashima residents, while maintaining that Japanese and South Korean views of the issue remained sharply opposed.[3]

## Current status

The island remains uninhabited, with empty high-rise buildings, seawalls and other mine-era structures exposed to wind and salt spray.[5] Contemporary travel writing and tourism media commonly present Hashima as an abandoned industrial landscape and a destination associated with heritage tourism, photography and urban exploration.[2][4] Its ruins have also become a subject of academic and creative work, including the AHRC-linked project "The Future of Ruins: Reclaiming Abandonment and Toxicity on Hashima Island" and the related 2013 performance "Return to Battleship Island".[3]

Hashima is widely circulated through popular culture and digital media, including fictionalized associations with "Skyfall" and survival-horror games.[4] A survey of immersive and virtual-reality projects found that such projects often emphasize either the former community life of the island or the visual spectacle of decay, while tending to avoid contested aspects of the site's history.[4] Educational and heritage-oriented digital resources have used Hashima to examine contested heritage, UNESCO inscription, national memory politics and commercial tourism.[4]

The island's interpretation continues to be the subject of diplomatic disagreement.[1] At the 47th session of the UNESCO World Heritage Committee in Paris, South Korea sought renewed examination of Japan's follow-up measures regarding Hashima, while Japan argued that it had implemented earlier resolutions.[1] The committee backed the position that disputes over Hashima should be addressed through bilateral dialogue rather than UNESCO procedures.[1] Japan's revised proposal was adopted by anonymous ballot, with seven votes in favor, three against and several abstentions among the 21 committee members.[1] Following that outcome, further formal discussion of the issue during the session, or in future committee meetings, was reported as unlikely.[1]

former Hashima residents, while maintaining that Japanese

# Root Cause Analysis

## - Why do Methods 1-3 generate Phantom Citations?

**Over-Constrained Prompting:** The system prompt aggressively forces the model to cite every sentence (e.g., "*You MUST cite your sources*"). To comply, the model **hallucinates citation tags** for general knowledge or transitional sentences where no actual source was retrieved.

**Wrong indexing:** Refer to the chunk index, instead of the URL index

### **Result:**

This artificially inflates the Citation Rate while severely degrading Citation Link Precision.

# Method 3 NLI

**What it is:** NLI is a core NLP task that evaluates the logical relationship between two pieces of text. It categorizes how Text A relates to Text B into three types:

- **Entailment:** Text A logically leads to or supports Text B. The entailment score specifically quantifies the model's confidence level that the first section acts as a logical prerequisite for the second.
- **Contradiction**
- **Neutral**

**Why we use it in Method 3 (The Engineering Value):**

- **Cost and Speed Optimization:** Having an LLM evaluate every possible section pair is too slow and consumes too many API tokens.
- **The "Fast-Track" Filter:** We use a lightweight NLI CrossEncoder model as the first gate in Agent 2. It quickly calculates a mathematical entailment score between section summaries. If the score is extremely high (above 1.5), the system instantly confirms a logical dependency and skips the LLM entirely.

# DAG (Directed Acyclic Graph)

**What it is:** A DAG is a fundamental mathematical data structure made of nodes connected by one-way arrows (directed edges). It has one absolute rule: **it must never contain a closed loop (a cycle)**. You can never follow the arrows and end up back where you started.

## **Why we use it in Method 3 (The Engineering Value):**

- **Preventing Logical Deadlocks:** In Agent 2, sections are nodes and "Prerequisite" dependencies are the arrows. If the system accidentally creates a paradox (e.g., Section A needs B, B needs C, and C needs A), the DAG structure allows the NetworkX algorithm to detect the loop and delete the weakest arrow to "break the cycle".
- **Enabling Topological Sorting:** A valid DAG can be flattened out mathematically. Topological sorting converts the web of connections into a strict, linear, 1-2-3 execution order. This guarantees that Agent 3 will always generate foundational background information before moving on to complex, derived topics, completely eliminating reverse causality.

# DAG (Directed Acyclic Graph)

## 1. PREREQUISITE (The Structural Backbone)

- The system draws a strict, one-way arrow from A to B. These arrows define the core graph. If they accidentally create a logic loop, the system steps in here to break the cycle.

## 2. RELATED (Dynamic Allocation)

- **How it is handled:** "Related" means the sections share context and shouldn't contradict each other, but they lack a strict "must-read-first" order. The DAG handles this dynamically rather than forcing an immediate directional edge.
- **DAG Impact:**
  1. **Hold & Wait:** The system records the pair in a temporary list and ignores them while building the initial DAG and resolving cycles.
  2. **Sort First:** It performs the topological sort based *only* on the PREREQUISITE edges to get the foundational execution sequence (`final_order`).
  3. **Dynamic Assignment:** Once the layout is locked, the system checks where the two related sections landed in the order. If the sorting placed Section A before Section B, the system seamlessly converts the "Related" tie into a one-way dependency (pushing A into B's dependency list).